Machine Learning Engineer

Job description

About GitHub

GitHub is the world’s leading platform for agentic software development — powered by Copilot to build, scale, and deliver secure software. Over 180 million developers, including more than 90% of the Fortune 100 companies, use GitHub to collaborate, and more than 77,000 organisations have adopted GitHub Copilot.

Locations

In this role you can work from Remote, Ontario Canada

Overview

GitHub is changing the way the world builds software and we want you to help build and secure GitHub. We're looking for an experienced machine learning engineer to help design, build and deploy agentic solutions, and to conduct ad-hoc analysis, as you help protect the home of all developers.

You will be responsible for identifying new trends relating to safety, fraud and abuse on GitHub, building agentic solutions to detect this abuse at scale, identifying vulnerabilities in GitHub that lead to abuse and helping to measure the impact of our work to safeguard the platform. At GitHub, Safety and Integrity's mission is to ensure GitHub and our users' safety through fighting malware, spam and fraud, monitoring for fake accounts, countering inauthentic content, battling crypto mining, and other core areas. You will be involved in collaborations across teams within GitHub including with Copilot and setting the standard for effective and responsible use of AI for moderation and trust and safety purposes, ensuring fraud is countered, content is moderated, users are kept safe and the open-source community can flourish.

If you have a strong foundation in large language models, solid software engineering instincts, a working knowledge of online platform trust and safety issues, and an empathetic approach to collaborating with a diverse team from entry-level associates to seasoned senior contributors, then this might be the gig for you.

What We Value

• Collaboration: We believe the best work is done together.
• Empathy: We believe in putting people first.
• Quality: We believe in setting the standard for excellence.
• Positive Impact: We believe in making the world a better place through our work.
• Shipping: We believe in creating things for the people using them.

Responsibilities

• Design, build and deploy agentic solutions that leverage large language models to detect and prevent fraud, abuse, and security threats at scale — applying LLMs to problems such as content classification and multi-step agentic investigation.
• Build well-engineered, production-grade systems that run reliably against high-volume event streams, making effective use of AI coding assistants to accelerate and improve your work.
• Build and operate scalable ML systems on cloud platforms (such as Azure AI Foundry) for training, deploying, and serving models and agentic solutions in production.
• Evaluate and improve existing models and agentic solutions using offline evaluations (including tool-use loops and LLM-as-judge evaluation), performance metrics, and feedback from operational deployments.
• Identify vulnerabilities in products that lead to abuse, and provide consultation to product teams reviewing new features.
• Collaborate closely with cross-functional teams including data scientists, software engineers, product managers and content moderators to integrate agentic solutions into production systems.
• Document the systems you help build and support the technical growth of your peers.

Qualifications

Required Qualifications

• 4+ years experience in machine learning, or related field
• OR Bachelor's Degree in Computer Science, Software Development, Electrical or Computer Engineering, Mathematical Sciences, or related field AND 2+ years experience in machine learning, or related field
• OR Master's Degree in Machine Learning, Computer Science, Software Development, Electrical or Computer Engineering, Mathematical Sciences, or related field
• OR equivalent experience.

Preferred Qualifications

• Strong understanding of large language models — how they work — and hands-on experience applying them at scale, ideally for classification, agentic workflows, or agents.
• Strong software engineering skills, including experience building with AI coding assistants.
• Experience designing or evaluating agentic systems (tool-use loops, multi-step workflows, or LLM-as-judge evaluation).
• Hands-on experience building and operating classification or detection systems at scale, including handling imbalanced data and precision/recall tradeoffs.
• Experience in Trust and Safety, National Security or fighting spam, malware, fraud, and threat actor activity at scale.
• Experience in responsible AI.
• Experience in Safety-by-Design.
• Experience with managing user data and privacy.
• Solid understanding of machine learning algorithms (supervised and unsupervised learning, anomaly detection, etc.) and their practical implementation.

GitHub values

• Customer-obsessed
• Ship to learn
• Growth mindset
• Own the outcome
• Better together
• Diverse and inclusive

Manager fundamentals

• Model
• Coach
• Care

Leadership principles

• Create clarity
• Generate energy
• Deliver success

Who We Are

GitHub is the world's leading AI-powered developer platform with 150 million developers and counting. We're also home to the biggest open-source community on earth (and 99% of the world's software has open-source code in its DNA). Many of the apps and programs you use every day are built on GitHub.

Our teams are dreamers, doers, and pioneers, leading the way in AI, driving humanitarian efforts around the globe, and even sending open source to Mars (and beyond!).
At GitHub, our goal is to create the space you need to do your best work. We're remote-first and offer competitive pay, generous learning and growth opportunities, and excellent benefits to support you, wherever you are—because we know that people flourish when they can work on their own terms.

Join us, and let's change the world, together.

Equal Employment Opportunity